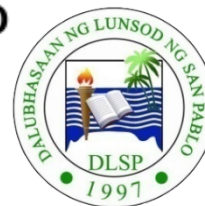




DALUBHASAAN NG LUNSOD NG SAN PABLO

COLLEGE OF COMPUTER STUDIES AND TECHNOLOGY
MIDTERM EXAMINATION IN ITELECTV 1-
OBJECT-ORIENTED PROGRAMMING
FIRST SEMESTER AY 2026-2027



Patriotism • Leadership • Service • Professionalism

NAME: _____

SCORE : _____

PROGRAM/SECTION: _____

INSTRUCTOR: _____

TEST I. MULTIPLE CHOICE. Choose the best answer for the following questions. Write the letter of the correct answer.

- What is programming?
A. The process of physically assembling computer hardware
B. The process of creating instructions that a computer can execute
C. The process of connecting computers to the Internet
D. The process of installing an operating system
- What is the primary purpose of a programming language?
A. To replace computer hardware
B. To provide a structured way for humans to communicate instructions to computers
C. To increase the storage capacity of a computer
D. To allow computers to understand human speech automatically
- What is an algorithm?
A. A set of computer hardware components
B. A file used to store program data
C. A step-by-step procedure for solving a problem
D. A tool used to design computer graphics
- Which Java component executes Java bytecode?
A. JDK
B. JVM
C. IDE
D. Javadoc
- Java's principle of "Write Once, Run Anywhere" is mainly possible because Java programs are compiled into:
A. machine code that works on every processor
B. Java source code
C. bytecode that can be executed by a compatible JVM
D. HTML code
- Which is the correct declaration of an integer variable in Java?
A. integer age = 18;
B. int age = 18;
C. age int = 18;
D. number age = 18;
- Which primitive data type is appropriate for storing true or false?
A. String
B. char
C. boolean
D. int
- Which value is within the valid range of Java's byte data type?
A. 200
B. -150
C. 127
D. 500
- Which statement correctly describes char and String?
A. Both are primitive data types.
B. char uses single quotes, while String normally uses double quotes.
C. String can contain only one character.
D. char uses double quotes, while String uses single quotes.
- What is the result of the following expression? $\text{int result} = 10 + 4 * 2;$
A. 28
B. 18
C. 24
D. 16
- Which operator checks whether two primitive values are equal?
A. =
B. ==
C. !=
D. :=



12. What will the following code display?

```
int age = 18;

if (age >= 18) {
    System.out.println("Adult");
} else {
    System.out.println("Minor");
}
```

- A. Minor
- B. Adult

- C. 18
- D. Nothing

13. Which loop guarantees that its body executes **at least once**?

- A. for
- B. while

- C. do-while
- D. enhanced for

14. In Java, what is a method?

- A. A variable that stores an object
- B. A block of code designed to perform a particular task

- C. A package containing Java classes
- D. A data type used only for numbers

15. In the following method, what are a and b?

```
public static int add(int a, int b) {
    return a + b;
}
```

- A. Arguments
- B. Classes

- C. Parameters
- D. Object

16. What does the return statement do in the method below?

```
public static int square(int number) {
    return number * number;
}
```

- A. Stops the entire Java application
- B. Sends a value back to the method caller

- C. Prints the value automatically
- D. Creates a new object

17. Which demonstrates valid **method overloading**?

- A. void show(int x) { } ; void show(String x) { }
- B. int show(int x) { return x; }; double show(int x) { return x; }

- C. void show(int x) { }; void display(int x) { }
- D. void show(int x) { }; void show(int x) { }

18. Given the following array, what is the value of scores[1]? int[] scores = {80, 90, 100};

- A. 80
- B. 90

- C. 100
- D. 1

19. What does scores.length return for the array below? int[] scores = {70, 80, 90, 100};

- A. 3
- B. 4

- C. 5
- D. 100

20. Which is the correct way to compare the **contents** of two String objects?

- A. text1 = text2
- B. text1 == text2

- C. text1.equals(text2)
- D. text1.compare = text2

21. Which statement best describes a **class**?

- A. A specific instance stored in memory
- B. A blueprint or programmer-defined reference type used to create objects

- C. A method that returns no value
- D. A primitive data type

22. What is an **object**?

- A. An instance of a class
- B. A Java package

- C. A primitive variable
- D. A loop statement

23. Consider: Student student1 = new Student(); What does new Student() do?

- A. Deletes a Student object
- B. Creates a new Student object and invokes its constructor

- C. Creates a new primitive type
- D. Imports the Student class

24. In a class, fields or attributes primarily represent an object's:

- A. loops
- B. state or data

- C. package name
- D. compiler

25. What happens in this code?
- ```
Student student1 = new Student();
Student student2 = student1;
```
- A. Two independent Student objects are created.  
B. student1 is deleted.  
C. Both variables refer to the same Student object.  
D. Java produces a compilation error.
26. What does null mean for an object reference?
- A. The object contains zero  
B. The reference currently refers to no object  
C. The object has been converted to a String  
D. The reference points to every object
27. What is the purpose of this in the following code?
- ```
public void setName(String name) {
    this.name = name;
}
```
- A. It refers to the current object's name field.
B. It creates another object.
C. It makes "name" static.
D. It refers to the superclass.
28. Which statement about a Java constructor is correct?
- A. A constructor must return void.
B. A constructor has the same name as its class and has no return type.
C. A constructor must always be static.
D. A constructor can only initialize static fields.
29. Which is a valid constructor for a class named Student?
- A. public void Student() { }
B. public student() { }
C. public Student() { }
D. public int Student() { return 0; }
30. When does Java provide a compiler-generated default constructor?
- A. Every time a class is created
B. Only when the class declares no constructors
C. Only when all fields are private
D. Only when the class contains main()
31. What concept is demonstrated below?
- ```
public Student() {
}

public Student(String name) {
 this.name = name;
}
```
- A. Method overriding  
B. Constructor overloading  
C. Inheritance  
D. Encapsulation
32. In a constructor, what is the purpose of this(...)?
- A. It invokes another constructor in the same class.  
B. It invokes a superclass constructor.  
C. It deletes the current object.  
D. It converts an instance method to static.
33. Which statement about static variables is correct?
- A. Every object receives an independent copy.  
B. They are associated with the class rather than each individual instance.  
C. They can never be changed.  
D. They can only store integers.
34. Why is Java's main() method declared static?
- A. So it cannot execute more than once  
B. So it can only access private variables  
C. So the Java launcher can invoke it without first creating an instance of the class  
D. So it automatically creates all objects
35. Which access modifier provides the most restrictive access for a field?
- A. public  
B. protected  
C. private  
D. static
36. What access level does a class member have when no explicit access modifier is written?
- A. public  
B. package-private  
C. protected  
D. private
37. Why are object fields commonly declared private?
- A. To prevent the class itself from accessing them  
B. To protect internal state from uncontrolled direct access  
C. To automatically make them constants  
D. To make them static
38. What is the primary purpose of a **getter** method?
- A. To create an object  
B. To provide controlled access for reading a value  
C. To make a field public  
D. To delete a field

39. Why is the if statement useful?

```
private int age;

public void setAge(int age) {
 if (age >= 0) {
 this.age = age;
 }
}
```

- A. It makes age static.
- B. It validates the value before changing the object's state.
- C. It converts age into a String.
- D. It makes the setter private.

40. Why is there no need to make balance public?

```
public class BankAccount {
 private double balance;

 public void deposit(double amount) {
 if (amount > 0) {
 balance += amount;
 }
 }

 public double getBalance() {
 return balance;
 }
}
```

- A. Because balance can never change
- B. Because deposit() and getBalance() provide controlled ways to interact with the balance
- C. Because private fields automatically become static
- D. Because methods cannot access private fields

TEST II – IDENTIFICATION. Write the correct word or term that best answers the question.

41. A step-by-step procedure or sequence of instructions designed to solve a particular problem.
42. A programmer-defined reference type that serves as a blueprint for creating objects.
43. A special member of a class that is invoked when an object is created and is commonly used to initialize the object's state.
44. A keyword that refers to the current object and can be used to distinguish an object's field from a parameter with the same name.
45. An access modifier that restricts a member so that it cannot be directly accessed by ordinary code outside its declaring class.

TEST III – ABBREVIATION. Write the complete meaning of each abbreviation

46. JVM - \_\_\_\_\_
47. JDK - \_\_\_\_\_
48. JRE - \_\_\_\_\_
49. OOP - \_\_\_\_\_
50. WORA - \_\_\_\_\_

|                        |                          |                         |                                                  |
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Answer key:

|     |   |     |                             |
|-----|---|-----|-----------------------------|
| 1.  | B | 26. | B                           |
| 2.  | B | 27. | A                           |
| 3.  | C | 28. | B                           |
| 4.  | B | 29. | C                           |
| 5.  | C | 30. | B                           |
| 6.  | B | 31. | B                           |
| 7.  | C | 32. | A                           |
| 8.  | C | 33. | B                           |
| 9.  | B | 34. | C                           |
| 10. | B | 35. | C                           |
| 11. | B | 36. | B                           |
| 12. | B | 37. | B                           |
| 13. | C | 38. | B                           |
| 14. | B | 39. | B                           |
| 15. | C | 40. | B                           |
| 16. | B | 41. | Algorithm                   |
| 17. | A | 42. | Class                       |
| 18. | B | 43. | Constructor                 |
| 19. | B | 44. | this                        |
| 20. | C | 45. | private                     |
| 21. | B | 46. | Java Virtual Machine        |
| 22. | A | 47. | Java Development Kit        |
| 23. | B | 48. | Java Runtime Environment    |
| 24. | B | 49. | Object-Oriented Programming |
| 25. | C | 50. | Write Once, Run Anywhere    |

# Two-Way TOS

Course: Object-Oriented Programming – Java

Assessment: Midterm Examination

Coverage: Weeks 1–6

Total Number of Items: 50

Total Points: 50

| Content Area                                 | Remembering                 | Understanding | Applying  | Analyzing | Total Items | Weight |
|----------------------------------------------|-----------------------------|---------------|-----------|-----------|-------------|--------|
| Week 1 – Introduction to Programming & Java  | 1, 3, 4, 41, 46, 47, 48, 50 | 2, 5          | —         | —         | 10          | 20%    |
| Week 2 – Java Programming Fundamentals       | 7, 13                       | 6, 9, 11      | 8, 10, 12 | —         | 8           | 16%    |
| Week 3 – Methods, Arrays, Strings & Packages | 14, 15                      | 16, 20        | 18, 19    | 17        | 7           | 14%    |
| Week 4 – Classes and Objects                 | 21, 22, 24, 42, 44, 49      | 26, 27        | 23        | 25        | 10          | 20%    |
| Week 5 – Constructors and Class Members      | 28, 30, 43                  | 31, 33, 34    | 29, 32    | —         | 8           | 16%    |
| Week 6 – Access Modifiers and Class Design   | 35, 36, 45                  | 37, 38        | 39        | 40        | 7           | 14%    |
| TOTAL                                        | 24                          | 14            | 9         | 3         | 50          | 100%   |
| Percentage                                   | 48%                         | 28%           | 18%       | 6%        | 100%        |        |

## Cognitive-Level Distribution

| Bloom's Cognitive Level | No. of Items | Percentage |
|-------------------------|--------------|------------|
| Remembering             | 24           | 48%        |
| Understanding           | 14           | 28%        |
| Applying                | 9            | 18%        |



|           |    |      |
|-----------|----|------|
| Analyzing | 3  | 6%   |
| Total     | 50 | 100% |

Test Format Distribution

| Test Type                | Item Numbers | No. of Items | Percentage |
|--------------------------|--------------|--------------|------------|
| Multiple Choice          | 1–40         | 40           | 80%        |
| Identification           | 41–45        | 5            | 10%        |
| Meaning of Abbreviations | 46–50        | 5            | 10%        |
| Total                    | 1–50         | 50           | 100%       |

Content Distribution

| Week   | Topic                               | No. of Items | Percentage |
|--------|-------------------------------------|--------------|------------|
| Week 1 | Introduction to Programming & Java  | 10           | 20%        |
| Week 2 | Java Programming Fundamentals       | 8            | 16%        |
| Week 3 | Methods, Arrays, Strings & Packages | 7            | 14%        |
| Week 4 | Classes and Objects                 | 10           | 20%        |
| Week 5 | Constructors and Class Members      | 8            | 16%        |
| Week 6 | Access Modifiers and Class Design   | 7            | 14%        |
|        | TOTAL                               | 50           | 100%       |